

Data Mining Assignment 1

Pre-Processing, Association Analysis, and Prediction Summary

Verified run generated on 2026-04-02

Run configuration

Setting	Value / Decision
Raw records	414
Population STD	Yes
Correlation method	Pearson
Correlation threshold	0.70
Confidence threshold	0.70
Random seed	100
10% split rule	Round up to guarantee at least 10% from each treat class
Final minimum support count used	17
Support-count note	Support 20 was tested, but it produced 0 surviving Format-1 rules with a treat conclusion. Support 17 was the strongest verified cutoff that still preserved treat-prediction rules for Part 3.
Prediction note	If no Format-1 rule matched a test record, the run defaulted to predict treat = 0 because the surviving final rules only concluded treat = 1.

1. Part 1 results

After the population-standard-deviation outlier scan, 105 records were removed. The remaining cleaned dataset contained 309 records.

Attribute	AVG	STD	Outlier Count
debt	44.037681	29.455846	23
lopen	101.026329	57.142082	19
lrer	98.002899	10.658237	30
bndvol	0.691546	0.662633	17
gdpgvol	1.837198	1.645655	21
pivol	0.995169	1.001437	15
spreadvol	0.905314	0.790475	14

The full outlier-value list and the removed-record detail file are included in the CSV package.

Pearson correlation matrix

Attribute	debt	lopen	lrer	bndvol	gdpgvol	pivol	spreadvol
debt	1	0.025	0.051	0.166	-0.04	0.006	-0.046
lopen	0.025	1	-0.157	0.106	0.149	0.195	-0.092
lrer	0.051	-0.157	1	-0.165	0.099	-0.046	0.057
bndvol	0.166	0.106	-0.165	1	0.048	0.279	-0.072
gdpgvol	-0.04	0.149	0.099	0.048	1	0.341	0.5
pivol	0.006	0.195	-0.046	0.279	0.341	1	0.235
spreadvol	-0.046	-0.092	0.057	-0.072	0.5	0.235	1

No absolute Pearson correlation coefficient exceeded the 0.70 threshold, so no redundant attribute was removed.

Entropy-based discretization summary

Attribute	Best Threshold	Best Information Gain	Diff_Union Candidate Count
debt	14.8	0.005666	3
lopen	155.9	0.060276	3
lrer	110.7	0.006476	3
bndvol	1.7	0.011479	18
gdpgvol	4.6	0.005495	4
pivol	0.6	0.017165	26
spreadvol	0.1	0.016632	24

The full Diff_Union candidate lists and the threshold-by-threshold information-gain calculations are included in the CSV package.

2. Part 2 results

The split was performed only after the Economy dataset had been fully discretized. Seed 100 makes the class-stratified random draw reproducible. The holdout counts were rounded up so Econ-TEST contains at least 10% of both treat groups.

treat	Discretized Class Count	10% Raw Value	Econ-TEST Sample Count
0	137	13.7	14
1	172	17.2	18

Table A

Attribute Name	Discrete Value	Naming	Range
treat	0	I1	0
treat	1	I2	1
debt	1	I3	[0.4-14.8)

debt	2	I4	[14.8-101.5]
lopen	1	I5	[44.7-155.9]
lopen	2	I6	[155.9-175.1]
lrer	1	I7	[77.2-110.7]
lrer	2	I8	[110.7-117.9]
bndvol	1	I9	[0.0-1.7]
bndvol	2	I10	[1.7-1.8]
gdpgvol	1	I11	[0.2-4.6]
gdpgvol	2	I12	[4.6-5.0]
pivol	1	I13	[0.0-0.6]
pivol	2	I14	[0.6-2.9]
spreadvol	1	I15	[0.0-0.1]
spreadvol	2	I16	[0.1-2.4]

Support-count verification

Minimum Support Count	Total Frequent Itemsets	Frequent 3-Itemsets	Maximum Frequent Itemset Length	Surviving Format-1 Rules with treat Conclusion
15	751	143	8	24
16	735	139	8	16
17	719	135	8	8
18	687	128	8	0
19	671	124	8	0
20	655	123	8	0

The end-to-end run showed that support counts 18, 19, and 20 all eliminated the final treat-prediction rules. For that reason, the completed assignment run uses minimum support count = 17.

Frequent itemset counts by level

Itemset Length	Frequent Itemset Count
1	12
2	55
3	135
4	199
5	182
6	101
7	31
8	4

Balance check

Econ-TRAIN treat = 1 Count	Econ-TRAIN treat = 0 Count	IR	Dataset Status	Rule Score Used
154	123	0.111913	Balanced	Confidence

No frequent itemsets had to be removed for containing more than one value from the same attribute. The removed-invalid-itemsets file is empty except for its header row.

Final Format-1-Rules used for prediction

Format-1
lopen = 2 $\wedge$ gdpgvol = 1 $\Rightarrow$ treat = 1 (confidence = 100.00%)
lopen = 2 $\wedge$ lrer = 1 $\wedge$ gdpgvol = 1 $\Rightarrow$ treat = 1 (confidence = 100.00%)
lopen = 2 $\wedge$ lrer = 1 $\wedge$ bndvol = 1 $\wedge$ gdpgvol = 1 $\Rightarrow$ treat = 1 (confidence = 100.00%)
lopen = 2 $\wedge$ bndvol = 1 $\wedge$ gdpgvol = 1 $\Rightarrow$ treat = 1 (confidence = 100.00%)
lopen = 2 $\Rightarrow$ treat = 1 (confidence = 100.00%)
lopen = 2 $\wedge$ lrer = 1 $\Rightarrow$ treat = 1 (confidence = 100.00%)
lopen = 2 $\wedge$ lrer = 1 $\wedge$ bndvol = 1 $\Rightarrow$ treat = 1 (confidence = 100.00%)
lopen = 2 $\wedge$ bndvol = 1 $\Rightarrow$ treat = 1 (confidence = 100.00%)

All 8 final Format-1-Rules conclude treat = 1 with 100% confidence. The full Format-1 and Format-2 lists are included in the CSV package.

3. Part 3 results

Prediction was performed on Econ-TEST one record at a time. Because the final rules only concluded treat = 1, unmatched records were documented and defaulted to predict treat = 0 so every test record still received one binary prediction.

2 x 2 prediction matrix

Cell		Count	
Actual 1 / Predicted 1 (TP)		4	
Actual 1 / Predicted 0 (FN)		14	
Actual 0 / Predicted 1 (FP)		0	
Actual 0 / Predicted 0 (TN)		14	
Metric	Value		Calculation
Accuracy	0.5625		$(4 + 14) / (4 + 14 + 0 + 14)$
Recall	0.2222		$4 / (4 + 14)$
Precision	1.0000		$4 / (4 + 0)$

4. Defense notes

1. Open with the sequence: outliers, correlation, discretization, split, Apriori, rules, balance check, prediction, and metrics.
2. Explain the seed clearly. Seed 100 freezes the entire class-stratified random draw, not a single row.
3. Explain the rounding choice. 17.2 was rounded up to 18 for treat = 1, and 13.7 was rounded up to 14 for treat = 0, so the holdout contains at least 10% of both classes.
4. Explain the support-count choice honestly. Support 20 was tested, but it eliminated the final treat-prediction rules. Support 17 was the strongest verified cutoff that still preserved them.
5. Explain Table A. It maps each attribute and discrete value to its I-label and range so the mined rules can be translated back into readable statements.
6. Explain the balance result. IR = 0.1119, which is below 0.4, so the training dataset was treated as balanced and standard confidence was used instead of Kulc.
7. Explain the prediction assumption. All final rules predicted treat = 1. To make one binary prediction per test record, unmatched rows were documented and defaulted to treat = 0.
8. Explain the metric pattern. The positive predictions were perfectly precise, but recall was low because only 4 of the 18 positive test rows matched the final rule set.

5. Package contents

The downloadable package includes the reusable Python script plus CSV files for every major calculation block: Part 1 statistics, removed records, correlation matrix, Diff_Union candidates, threshold evaluations, discretized Economy, split tables, Table A, frequent itemsets, surviving rules, prediction rows, and final metrics.